ENERGIA: USB-C, carga de bateria, reparto de carga y regulador 3.3 V

J1 — USB-C 16P (VBUS, CC1, CC2, USB_DM, USB_DP, SBU1, SBU2, SHIELD, GND)

U1 — TP4056 (CE, CHRG, STDBY, TEMP, PROG, BAT, VCC, EPAD, GND)

J2 — BAT+ / BAT−

Bateria LiPo soldada a pads.
Usar una con proteccion integrada

Q1 — AO3401A (VBUS, VBAT, VSYS)

U2 — AP2112K-3.3 (VSYS, PWR_EN, VIN, EN, VOUT, GND, 3V3_REG)

D3 — B5819W (VSYS, VBUS)

SW1 — ENCENDIDO (PWR_EN, VSYS, GND)

JP1 — 3V3_REG / +3V3 / 3V3

R1 5.1k CC1 · R2 5.1k CC2 · C1 10uF VBUS · R3 2k PROG · C2 10uF VBAT · R6 100k VBUS · C3 1uF VSYS · C4 10uF 3V3_REG · R15 100k PWR_EN · R11 1M VBAT · R12 1M BAT_SENSE · C11 100nF BAT_SENSE · R4 1k VBUS · R5 1k VBUS

D1 — CHRG_N / LED_CHG_A — CARGANDO (rojo)

D2 — STDBY_N / LED_FULL_A — CARGA COMPLETA (verde)

Puntos de prueba y banderas de alimentacion

TP1 VBUS · TP2 VBAT · TP3 VSYS · TP4 +3V3 · TP5 GND · TP6 EN · TP7 BTN_MENU · TP8 UART_TX · TP9 UART_RX

PWR_FLAG VBUS · PWR_FLAG VSYS · PWR_FLAG +3V3 · PWR_FLAG GND

PANTALLA (conector PROVISIONAL)

J3 — LCD 2.4in ST7789V (+3V3, GND, LCD_SCK, LCD_MOSI, LCD_CS, LCD_DC, LCD_RST, LCD_LEDA, LCD_LEDK)

Q2 — AO3400A (LCD_BL, LCD_LEDK, GND)

R13 10R (por definir) +3V3 / LCD_LEDA

R14 100k LCD_BL / GND

Pines y resistencia de luz de fondo:
se definen al tener el panel en la mano

microSD (SPI)

J4 — TF-01A (GND, GND, SD_MISO, SD_CS, SD_MOSI, SD_SCK, DAT0(DO), DAT1(RSV), DAT2(RSV), CDDAT3(CS), CMD(DI), CLK(SCLK), VDD, CD, VSS, GND, GND)

R8 10k +3V3 / SD_CS · R9 10k +3V3 / SD_MISO · C8 100nF +3V3 / GND · C12 10uF +3V3 / GND

ESP32-S3-WROOM-1-N16R8

U3 — ESP32-S3-WROOM-1-N16R8 (+3V3, EN, BTN_MENU, BAT_SENSE, BTN_L, BTN_UP, BTN_DOWN, BTN_LEFT, BTN_RIGHT, BTN_A, LCD_DC, LCD_CS, LCD_MOSI, LCD_SCK, LCD_BL, LCD_RST, SD_MOSI, SD_SCK, UART_TX, UART_RX, SD_MISO, SD_CS, USB_DM, USB_DP, BTN_B, BTN_R, BTN_OPTION, I2S_BCLK, I2S_LRCLK, I2S_DIN, BTN_START, BTN_SELECT, GND)

SW2 — RESET (EN, GND)

R7 10k +3V3 / EN · C5 1uF EN / GND · C6 10uF +3V3 / GND · C7 100nF +3V3 / GND

BOTONES (cada uno de su GPIO a GND, pull-up interno)

SW3 UP (BTN_UP, GND) · SW4 DOWN (BTN_DOWN, GND) · SW5 LEFT (BTN_LEFT, GND)

SW6 RIGHT (BTN_RIGHT, GND) · SW7 A (BTN_A, GND) · SW8 B (BTN_B, GND)

SW9 START (BTN_START, GND) · SW10 SELECT (BTN_SELECT, GND) · SW11 MENU (BTN_MENU, GND)

SW12 OPTION (BTN_OPTION, GND) · SW13 L (BTN_L, GND) · SW14 R (BTN_R, GND)

MENU (GPIO 0): presionado al encender entra en modo de grabacion.
Solo se usan dos pads en diagonal de cada boton; los otros dos quedan sin conectar.

AUDIO (I2S)

U4 — MAX98357A (VSYS, I2S_DIN, I2S_BCLK, I2S_LRCLK, AMP_SD, SD_MODE, GAIN_SLOT, GND, EP PAD, VDD, DIN, BCLK, LRCLK, OUTP, OUTN, SPK_P, SPK_N)

J5 — BOCINA+ / BOCINA− (SPK_P, SPK_N)

R10 1M VSYS / AMP_SD · C9 10uF VSYS / GND · C10 100nF VSYS / GND

GAIN sin conectar = 9 dB

Borrador generado. Sin probar en hardware.

Sheet: /	
File: magsafe-console.kicad_sch	
Title: MagSafe Console	
Size: A2	Date:
KiCad E.D.A. 9.0.9	**Rev: 0.1**
	Id: 1/1